

Domain-Transferred Synthetic Data Generation for Improving Monocular Depth Estimation

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Abstract: A major obstacle to the development of effective monocular depth estimation algorithms is the difficulty in obtaining high-quality metric depth data that corresponds to real-world RGB images. Collecting this data is time-consuming and costly, and even data collected by modern sensors has limited range or resolution, and is subject to inconsistencies and noise. Data generated in simulation avoids these problems with accurate depth information, but models trained on synthetic data often do not transfer well to real world applications. To combat this, we propose a method of data generation in simulation using 3D synthetic environments and CycleGAN domain transfer to increase the realism of simulated images. We analyze this data generation method by training multiple depth estimation models on different datasets, including synthetic and domain-transferred data. We evaluate the performance of the models on the NYU-Depth V2 dataset to verify the generalizability of the approach and show that GAN-transformed data effectively helps to bridge the gap between simulated and real-world data in depth estimation.

Keywords: Computer Vision, Machine Learning, Monocular Depth Estimation, Sim-to-Real Transfer

1. INTRODUCTION

Obstacle detection and avoidance is essential for autonomous systems, particularly in autonomous driving and robotics [1], [2]. These systems require accurate distance measurements between robots and nearby obstacles to ensure safe and effective control, and while Stereo cameras and LiDAR distance sensors are capable of obtaining depth information, their weight, energy consumption, and resolution limitations make them less applicable to small robots constrained by size and power capacities. Monocular cameras have gained prominence in the realm of Micro Aerial Vehicles (MAVs) and mobile robots due to their low cost, lightweight, and ability to capture high-resolution images [3], [4]. While monocular vision inherently lacks depth perception, developing effective methods to estimate distance solely from monocular camera images holds significant potential for a wide range of applications.

The most effective methods for monocular depth estimation have been driven by developments in deep learning algorithms [5], [6], [7]. Deep learning requires large training image sets and corresponding depth data. Collecting accurate depth information presents several challenges, including the low spatial resolution of LiDAR data and the limited range and sensitivity to noise of stereo cameras. Additionally, manual data collection is time-consuming and resource-intensive. A possible alternative can be found in game engines such as Unreal Engine or Unity, which are employed to create synthetic environments that mimic real-world conditions [8], [9]. These synthetic environments allow the generation of im-

ages and the corresponding depth values automatically, and provide perfect depth measurements that are not subject to noise.

While images generated by game engines may appear realistic to the naked eye, it has been well acknowledged that they may contain subtle artifacts such as lighting or shadow artifacts, leading to biases in distance estimations [10], [11]. To bridge this domain gap between synthetic and real world imagery, we propose leveraging Generative Adversarial Networks (GANs) [12] for domain transfer. By utilizing a type of GAN called “CycleGAN,” [13] we transform synthetic images generated in a simulated environment into images closely resembling real-world scenes. Our research uses the transformed images as input for depth estimation models, demonstrating that training with a transferred synthetic dataset effectively minimizes the disparity between synthetic and real-world data, thereby enhancing depth map accuracy. Additionally, the high resolution of the synthetic dataset, compared to the real dataset, can improve the performance of the trained models. Our contributions in this work are as follows.

- Proposed a data augmentation framework that incorporates CycleGAN-transformed synthetic images to mitigate the scarcity of labeled real-world depth data.
- Demonstrated improved performance of monocular depth estimation models when trained on the augmented dataset, highlighting enhanced cross-domain generalization.

2. RELATED WORK

2.1 Domain Transfer

Generative Adversarial Networks (GANs) have become a cornerstone in generating realistic synthetic im-

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ages through adversarial training between two neural networks: a generator and a discriminator. The generator creates synthetic images, aiming to fool the discriminator, while the discriminator learns to distinguish real from fake. This dynamic competition progressively improves both models.

Building on this concept, Cycle-Consistent GAN (CycleGAN) enable unsupervised image-to-image translation between two domains without requiring paired data. CycleGAN uses two sets of generators and discriminators—one for each domain—and introduces a cycle consistency loss. This ensures that an image translated to the other domain and then back retains its original characteristics, making CycleGAN highly effective for applications where aligned image pairs are unavailable.

CycleGAN-based domain adaptation has gained traction across various computer vision tasks due to its ability to improve model generalization across domains. Some earlier GAN-based approaches, such as the style transfer method in [14], adapted real images for depth estimation but struggled in high-contrast scenes, where intensity mismatches caused shadows to be misinterpreted as foreground objects. Similarly, Zheng et al. [15] proposed a GAN-based image translation and monocular depth estimation framework, but the lack of cycle consistency led to significant visual artifacts.

In contrast, recent studies leveraging CycleGAN have shown promising results. For instance [16] demonstrated that synthetic-to-real translation improved YOLO’s performance in multi-organ detection in CT images, while [17] applied real-to-virtual domain transfer in transnasal surgery to enable stable 3D visualization. Zhao et al. [18] introduced GASDA, which uses bidirectional CycleGAN translation to maintain geometric consistency and improve monocular depth estimation. Similarly [19] presented RDB-CycleGAN with Residual Dense Blocks and SSIM loss for realistic CAD part translation, and [20] demonstrated enhanced zero-shot drone detection by training on synthetic images translated to the real domain using CycleGAN.

Motivated by these findings, we hypothesize that CycleGAN is particularly well-suited for our application. Specifically, we aim to translate synthetic images into realistic counterparts for monocular depth estimation, with the goal of reducing domain shift while preserving essential geometric structures.

2.2 Monocular Depth Estimation

Monocular Depth Estimation (MDE) is a method for predicting depth information from a single RGB image. The characteristic that MDE only uses RGB cameras without any other sensor offers a cost-effective and lightweight solution for depth perception.

Supervised learning-based MDE models are trained using datasets that have RGB and corresponding grayscale depth map image pairs such as NYU-Depth V2 [21] and KITTI [22]. Eigen [6] first proposed a deep learning-based MDE model which uses scale-invariant error and has a coarse-to-fine framework, where the

coarse network trains the global depth information and the fine network obtains local features. Since then, various deep learning methods for monocular depth estimation have emerged [5], [7].

The most common architecture for monocular depth estimation based on deep learning is an encoder-decoder network, using RGB images as input and extracted depth maps as an output. The encoder network consists of convolutional and pooling layers to learn features of images and the decoder network consists of deep convolutional layers to predict depth maps [23]. Because of the difficulty of collecting ground truth depth data, several different approaches have been explored to improve either the efficiency of data usage or expand the type of training data that can be used. Ranftl et al. introduced MiDaS[24], where they suggested several methods to allow training on multiple types of depth datasets, such as proposing training objectives that are invariant to depth range and scale, and exploring multi-objective learning when combining datasets.

While MiDaS computed relative depth maps, it was not capable of metric depth estimation. ZoeDepth [25] added this capability to the base MiDaS model by combining MiDaS with a metric depth binning module appended to the decoder. Also, Marigold [26] adapts a pre-trained stable diffusion model for monocular depth estimation. It fine-tunes only the U-Net using affine-invariant normalization and synthetic RGB-D data.

While supervised methods have been widely explored, more recently unsupervised and semi-supervised methods have also begun to gain traction, mainly because of their ability to learn without the same level of dependence on ground truth depth data. Most notably, recent benchmarks have been set on the NYU-Depth V2 dataset by models such as VPD(Visual Perception with a pre-trained Diffusion model) [27] which leverages cross-attention maps between images and text labels to aid segmentation and depth estimation tasks, and Depth Anything [28] which uses combinations of ground truth labels and pseudo-labels generated in a teacher-student architecture to learn from both labeled and unlabeled image datasets. While both of these models mainly rely on large training image datasets that do not contain detailed depth information, they still require ground truth depth data for fine-tuning.

Similar to our work, the authors of [14] explore the use of domain transfer between real and synthetic data to aid in MDE. In their work, they use domain transfer to transform real domain images into simulated domain images, using CycleGAN as a part of the final model. However, this requires the domain transfer to be done in real-time, adding model complexity and requiring more processing power. Instead, we explore the reverse approach, by translating synthetic training data into the real domain in order to more effectively train MDE models without the need to incorporate CycleGAN as a part of the model itself.

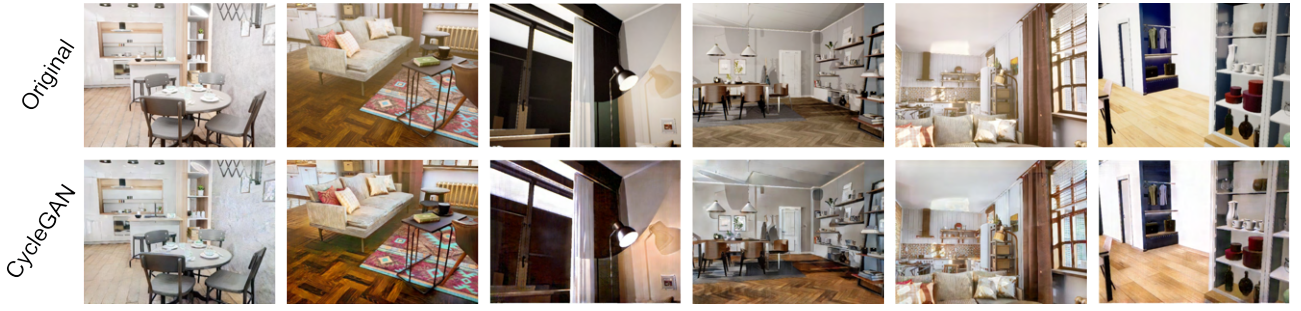


Fig. 1. Synthetic images generated using Unreal Engine (first row) and their CycleGAN-transformed counterparts (second row). The CycleGAN transformation process alters features such as lighting and texture to increase the realism of a synthetic image while maintaining the original features of the environment.

3. PROPOSED METHOD

Our proposed method consists of three modules: synthetic data generation, data domain transformation, and depth estimation model training.

3.1 Synthetic Data Generation

To generate the synthetic image dataset, we utilized the Unreal Engine v4.27 simulation software developed by Epic Games [8], in conjunction with the AirSim [29] for capturing RGB images and pixel-wise distance data. We selected a diverse set of indoor virtual environments, including apartment scenes, living room, bedroom, and kitchen environments. To match the image data in the NYU-Depth V2 dataset, the RGB images were captured at a resolution of 640x480 pixels, and the camera FOV was set to 57 degrees. The depth information was captured using the AirSim DepthPerspective camera setting and was saved to the .pfm file format to preserve raw distance information. This .pfm file was then converted to a 16-bit depth image by first truncating any distances exceeding 10 meters to a value of 10 meters and scaling the result depth values in millimeters as a 16-bit integer grayscale .png image to match the NYU-Depth V2 dataset.

3.2 Domain Transfer

CycleGAN’s ability to perform unpaired image-to-image translation is especially valuable for domain transfer tasks where paired training examples are scarce or unavailable. This aligns with our objective of translating synthetic images into the real domain without requiring one-to-one correspondences.

In our approach, we use CycleGAN to translate images generated in Unreal Engine into a more realistic style. The synthetic dataset is constructed using the methodology described in the previous section. For the real domain, we curate a subset of images from NYU-Depth V2, House Rooms Image [30], House Rooms [31], and Position Image [32] datasets, selecting scenes that visually align with the synthetic data.

The resulting translated dataset—produced by CycleGAN—is then used to train the monocular depth estimation models. Fig. 1 shows examples of Unreal Engine-generated synthetic images and their CycleGAN-translated counterparts.

3.3 Depth Estimation

We implement three recent monocular depth estimation models to evaluate the effectiveness of our image domain transformation approach across architectures. We chose to implement ZoeDepth [25], DepthAnything [28], and Marigold [26]. These models were selected because they represent a variety of different approaches to depth estimation, and if domain transfer is an effective method of improving synthetic depth data, we should observe consistent results across model types.

ZoeDepth employs the MiDaS [24] training strategy for relative depth estimation using a scale and shift invariant loss. After pretraining, it introduces a Metric Bins Module based on an adaptive binning principle [33] to the decoder, enabling fine-tuning for accurate metric depth estimation. DepthAnything expands on a ZoeDepth backbone but focuses on diverse pretraining on large amounts of relative depth data to improve generalization capabilities before fine-tuning on metric depth data. Marigold, in contrast, employs a pretrained latent diffusion model for monocular depth estimation, and fine-tunes for affine-invariant depth prediction using only a denoising U-Net.

To evaluate the effect of domain adaptation, each model is trained on both domain-transferred and non-domain-transferred simulated data, and evaluated on the NYU-Depth V2 dataset to assess real-world performance. This allows us to effectively evaluate the impact of domain-transfer on the final prediction results across a variety of models.

4. EXPERIMENTAL RESULTS

4.1 Dataset Composition and Pre-Processing

To determine the impact of different types of training data on the prediction accuracy of depth estimation models, we created two datasets of synthetic depth images. The first dataset, UE10k, consisted of 10k Unreal Engine synthetic images from seven different environments. The second dataset, GAN10k, consisted of the 10k synthetic images from the UE10k dataset, that had been translated to the real domain using CycleGAN. For validating the results of our method, we use the popular NYU-Depth v2 dataset for testing using the official test split.

Table 1. Error metrics for models trained using domain-transferred data (GAN10k) and non-domain-transferred data (UE10k) and tested on the NYU-Depth V2 Test set. The models trained on the GAN10k data perform better than those trained on the UE10k data, showing the effectiveness of domain-transfer in real-world generalization. Note that ZoeDepth and DepthAnything predict true metric depth, while Marigold predicts affine-invariant depth. This results in better metrics and smaller improvement margins from domain transfer for Marigold due to the simpler prediction task.

	NYU Test	$\delta 1$ ($\uparrow$)	$\delta 2$ ($\uparrow$)	$\delta 3$ ($\uparrow$)	AbsRel ($\downarrow$)	RMSE ($\downarrow$)	log10 ($\downarrow$)	RMSE_log ($\downarrow$)	SI_log ($\downarrow$)	SqRel ($\downarrow$)
ZoeDepth	UE10k	0.408	0.633	0.766	0.320	0.811	0.203	0.639	46.07	0.308
	GAN10k	0.406	0.642	0.777	0.313	0.803	0.195	0.606	43.02	0.297
DepthAnything	UE10k	0.440	0.654	0.761	0.321	0.812	0.212	0.706	55.43	0.317
	GAN10k	0.452	0.649	0.759	0.316	0.777	0.212	0.707	54.02	0.305
Marigold	UE10k	0.940	0.988	0.997	0.082	0.305	0.035	0.112	11.185	0.041
	GAN10k	0.943	0.988	0.997	0.079	0.306	0.034	0.109	10.850	0.041

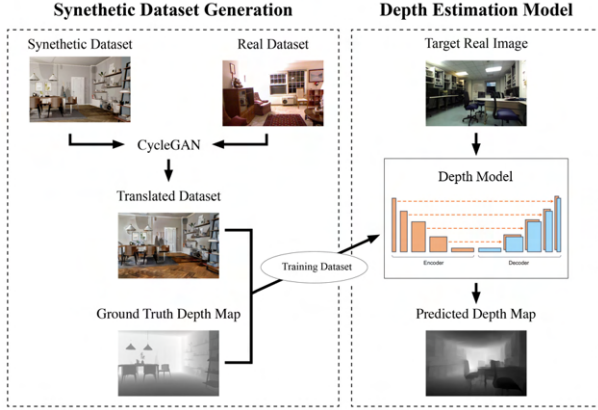


Fig. 2. CycleGAN-based domain transfer is applied to synthetic images generated from Unreal Engine to enhance realism. The resulting images are used for training a monocular depth estimation model.

4.2 Implementation Details

The three depth models (ZoeDepth, DepthAnything, and Marigold) were trained on an AMD Ryzen Threadripper 3960X 24-Core Processor and two NVIDIA RTX 3090s using the default parameters (batch size, optimizer, etc.) as specified in their original works [25], [28], [26]. All three models were trained for a total of 15 epochs.

For the domain transfer component of our proposed method, we implemented the CycleGAN framework as per the original specifications outlined in [13], [34]. The CycleGAN was trained with the same CPU and GPU used to train MDE model, and was trained for 200 epochs with learning rate of 0.0001.

4.3 Test Results

Table 1 shows the error results for the models trained on both the UE10k and GAN10k datasets and tested on the NYU-Depth v2 dataset. For evaluation metrics we use the $\delta 1$, $\delta 2$, and $\delta 3$ threshold accuracies, which measure the percentage of pixels within the threshold factor of the ground truth. We also use absolute relative error, root mean square error (RMSE), log10 error, RMSE log error, scale-invariant log error, and squared relative error, all of which are commonly used for evaluating MDE models.

Across the different models, both synthetic datasets had good performance, with the GAN10k dataset performing slightly better than the UE10k dataset on av-

erage, showing that domain transfer can help synthetic datasets to generalize to real world applications.

It is important to note that these results are zero-shot depth estimation, as none of the NYU test set was used as training data with the exception of for the CycleGAN. However, even with a relatively small dataset size and zero-shot testing, synthetic data yields effective depth estimation performance. While the performance gains from domain transfer are small, they are consistent across models, improving their generalization performance.

An additional note is that while ZoeDepth and DepthAnything predict true metric depth, Marigold predicts affine-invariant depth which leaves the predictions subject to an unknown scale and shift. This results in better overall metrics and smaller improvement margins for Marigold due to the simpler prediction task.

Fig. 3 shows the qualitative results for all three models trained on both the UE10k and GAN10k datasets when tested on the NYU-Depth v2 test set. While all the models produce effective depth predictions, the models trained on domain-transferred (GAN) data perform better than their non-transferred (UE) counterparts, most notably in areas such as the the headboard of the bed and the chair at the head of the table in the examples.

4.4 Discussion

While we showed that GAN-transformed data can serve as a suitable alternative to real-world data, there are several improvements that could be made to our approach. One factor that impacted the results of the CycleGAN model was the inclusion of translated images with artifacts in the dataset. Since CycleGAN transformation is imperfect, the results can sometimes contain artifacts that do not correspond to the original input image. Fig. 4 shows examples of artifacts that can occur in GAN-translated images. While these images in this work were kept to preserve dataset size and diversity, the artifacts likely limited the prediction performance of the models trained on the GAN data. As such, future work could benefit by defining criteria to filter out images with translation artifacts. Additionally, there are studies that address this limitation by incorporating semantic segmentation and depth information in the CycleGAN training to preserve the semantics of the translated scenes [18], [35], [36].

Additionally, the simulated environments used to gen-

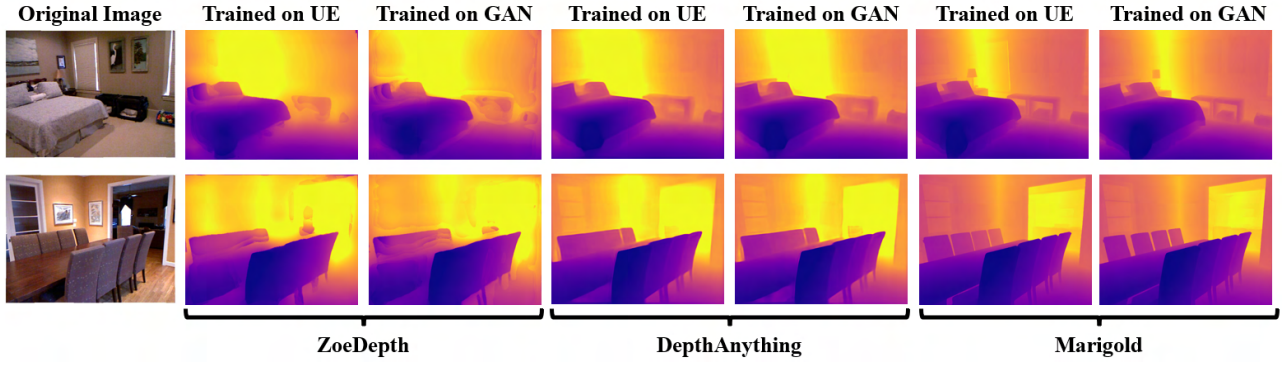


Fig. 3. Qualitative comparison between predicted depth maps for all three models. While the models perform well when trained on both the simulated (UE) and domain-transferred (GAN) data, the models trained on the GAN data more accurately predict areas such as the headboard of the bed and the chair at the head of the table.



Fig. 4. Examples of artifacts present in some CycleGAN-transformed images. While the overall image structure remains intact, some parts of the image, especially blank surfaces, can contain hallucinated textures or colors.

erate the synthetic data were mainly limited to indoor home environments, which could have constrained the results. Including more environments with a wider range of object details and lighting conditions could have a positive impact on the resulting models, and make them more robust to varying conditions.

5. CONCLUSION

In this work, we proposed a method to enhance performance of monocular depth estimation by increasing the realism of synthetic data using CycleGAN-based domain transfer. We generated depth images in a 3D simulation environment and applied CycleGAN-based domain transfer to enhance their realism, then compared the performance of multiple depth estimation models trained on this data. Through validation on the NYU-Depth v2 dataset, we verified the generalization performance of models trained on synthetic and domain-transferred data, and showed that using domain transfer improves the performance of models trained using synthetic data.

Synthetic data offers many advantages over real data, such as lower collection costs, higher resolution depth information, and unlimited range capability, but its lack of realism has been a major limitation. By applying CycleGAN domain transfer to help bridge that gap, our work demonstrated that synthetic data can serve as an effective resource for training effective monocular depth estimation models.

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